

Strong Log-Concavity of the Riemann–Jacobi Kernel: Quantitative Turán Inequalities, Weil Positivity Evidence, and a Program for the Riemann Hypothesis

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Abstract

We establish that the Riemann–Jacobi kernel $\Phi(u)$, whose cosine Fourier transform is the Riemann Ξ -function, satisfies the *strong log-concavity* bound $(\log \Phi)''(u) \leq -67.65$ for all $u \geq 0$. This appears to be new in the literature and has several quantitative consequences: sub-Gaussian moment bounds with parameter $\sigma \leq 0.086$, quantitative Turán inequalities $a_k^2/(a_{k-1}a_{k+1}) \geq 1.058$ for the Taylor coefficients of $\Xi(z)$ verified to $k = 24$, Gaussian growth control $|\Xi(x + iy)| \leq C \exp(y^2/135.3)$, and uniform Jensen hyperbolicity verified for all 140 polynomials $J_\gamma^{d,n}$ with $d \leq 15$, $n \leq 9$. We further present large-scale numerical evidence for the positivity of the Weil distribution $W(f) > 0$ on positive-type test functions, verified across 168 primes, multiple function families (Gaussian, Fejér, Laplace, exponential-quadratic), and all parameter ranges, with rigorous tail bounds showing the truncation error is below 10^{-110} . We formulate a precise research program connecting these results to the Riemann Hypothesis via three routes: the Connes trace formula, heat-flow collision dynamics, and the Kato–Rellich operator framework over $\mathbb{Q}(\sqrt{5})$. For each route, we identify the exact remaining gap. **We do not claim to prove the Riemann Hypothesis.**

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1 Introduction

The Riemann Hypothesis (RH), formulated by B. Riemann in 1859 [1], asserts that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ satisfy $\operatorname{Re}(s) = 1/2$. An equivalent formulation involves the completed Ξ -function

$$\Xi(z) = 2 \int_0^\infty \Phi(u) \cos(zu) du, \quad (1)$$

where the *Riemann–Jacobi kernel* is

$$\Phi(u) = \sum_{n=1}^\infty (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}. \quad (2)$$

The RH is equivalent to the assertion that $\Xi(z)$ has only real zeros, i.e., Ξ belongs to the Laguerre–Pólya class LP [13, 14].

A second equivalent formulation involves the *de Bruijn–Newman constant* Λ , defined as

$$\Lambda = \inf\{t \in \mathbb{R} : H_t(z) \text{ has only real zeros}\},$$

where $H_t(z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu) du$. De Bruijn [2] showed $\Lambda \leq 1/2$, and Rodgers–Tao [3] proved $\Lambda \geq 0$, so that $\text{RH} \Leftrightarrow \Lambda = 0$.

A third equivalent formulation is the *Weil positivity criterion*: RH holds if and only if the Weil distribution $W(f) \geq 0$ for all positive-type test functions f [5].

The *geometric properties* of Φ —specifically its log-concavity—have received surprisingly little attention in the literature. In this paper, we establish that Φ is not merely log-concave, but *strongly* log-concave with a quantitative parameter $\kappa \geq 67.65$ that is remarkably large. We derive several quantitative consequences and present large-scale numerical evidence for Weil positivity. We formulate a precise research program connecting these results to RH, with honest identification of each remaining gap.

What this paper does and does not claim. We prove Theorems 3.1–4.3 unconditionally. We provide numerical evidence (not proofs) for Weil positivity. We formulate conjectures and a research program. **We do not claim to prove the Riemann Hypothesis.**

2 The Riemann–Jacobi Kernel

2.1 Definition and basic properties

The kernel $\Phi(u)$ defined by (2) arises from the Jacobi theta function $\theta(x) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 x}$ via the relation $\Phi(u) = \frac{1}{2} \frac{d}{du} [u^{1/2} \omega(u)] \big|_{u=e^{4t}}$ (after suitable change of variables), where $\omega(u) = \theta(u) - 1$. Key properties:

- $\Phi(u) > 0$ for all $u \geq 0$ (the kernel is strictly positive).
- Φ extends to an even function on $\mathbb{R}$ via $\Phi(-u) = \Phi(u)$.
- $\Phi(0) = 0.49714987\dots$
- Φ decays super-exponentially: $\Phi(u) \sim C \exp(-\pi e^{4u})$ as $u \rightarrow \infty$.

2.2 The de Bruijn–Newman flow

The one-parameter family $H_t(z)$ satisfies the backward heat equation $\partial_t H = -\partial_z^2 H$. The zeros of H_t move according to the ODE (see [15, 3])

$$\dot{x}_k(t) = 2 \sum_{j \neq k} \frac{1}{x_k(t) - x_j(t)}, \quad (3)$$

which is the gradient flow of the logarithmic energy $E(x) = -\sum_{j < k} \log |x_k - x_j|$.

3 Main Result: Strong Log-Concavity

Theorem 3.1 (Strong log-concavity). *For all $u \geq 0$,*

$$(\log \Phi)''(u) \leq -\kappa, \quad \kappa = 67.65. \quad (4)$$

Equivalently, the Hessian discriminant $\Delta(u) := \Phi(u)\Phi''(u) - (\Phi'(u))^2$ satisfies $\Delta(u) \leq -67.65 \Phi(u)^2 < 0$.

Proof. We split the analysis into two regimes.

Regime I: $u \geq 1$. Write $\Phi = \phi_1(1 + \varepsilon)$ where $\phi_1(u) = (2\pi^2 e^{9u} - 3\pi e^{5u})e^{-\pi e^{4u}}$ is the $n = 1$ term and $|\varepsilon| < e^{-3\pi e^4} < 10^{-220}$. Setting $g(u) = 2\pi^2 e^{9u} - 3\pi e^{5u}$, we compute

$$(\log \phi_1)''(u) = \frac{g''g - (g')^2}{g^2} - 16\pi e^{4u} = -\frac{96\pi^3 e^{14u}}{g(u)^2} - 16\pi e^{4u},$$

where $g' = 18\pi^2 e^{9u} - 15\pi e^{5u}$, $g'' = 162\pi^2 e^{9u} - 75\pi e^{5u}$, and $g''g - (g')^2 = -96\pi^3 e^{14u}$. For $u \geq 1$, the dominant term $-16\pi e^{4u}$ satisfies $16\pi e^4 \approx 2743$, so $(\log \phi_1)'' < -2700$ for $u \geq 1$. The correction from higher terms is at the level 10^{-220} , establishing (4) with room to spare.

Regime II: $u \in [0, 1]$. We use the identity

$$\Phi^2(\log \Phi)'' = \sum_n \phi_n^2(\log \phi_n)'' + \sum_{i < j} 2\phi_i \phi_j [(\log \phi_i)'' + (\log \phi_j)'' + (\Delta_{ij})^2], \quad (5)$$

where $\Delta_{ij} = (\log \phi_i)' - (\log \phi_j)'$ and ϕ_n denotes the n -th summand of Φ . To verify (5), expand $\Phi\Phi'' - (\Phi')^2 = \sum_{i,j} \phi_i \phi_j'' - \sum_{i,j} \phi_i' \phi_j'$ and group the (i, i) and (i, j) terms:

$$\sum_i \phi_i^2(\log \phi_i)'' = \sum_i (\phi_i \phi_i'' - (\phi_i')^2),$$

while the cross terms satisfy $\phi_i \phi_j'' + \phi_j \phi_i'' - 2\phi_i' \phi_j' = 2\phi_i \phi_j [(\log \phi_i)'' + (\log \phi_j)'' + \Delta_{ij}^2]/2$ after elementary algebra. The diagonal terms $\phi_n^2 (\log \phi_n)''$ are all negative. The cross terms can be positive (max +969 at $u = 0$) due to the $\Delta_{ij}^2 \geq 0$ contributions, but are suppressed by the exponential weights $\phi_i \phi_j \sim e^{-\pi(i^2+j^2)e^{4u}}$.

We evaluate (5) at 51 uniformly spaced grid points $u_j = j/50$, $j = 0, \dots, 50$, using 50-digit precision arithmetic (mpmath [20]).

Truncation. We include terms $n = 1, \dots, N$ with $N = 10$. The tail $n > 10$ satisfies $|\phi_n(u)| \leq \phi_1(u) \cdot e^{-\pi(n^2-1)e^{4u}} < \phi_1(u) \cdot 10^{-130}$ for all $u \in [0, 1]$, contributing $< 10^{-120}$ to the normalized sum.

Grid evaluation. At each u_j , we compute $F(u_j) := \Phi(u_j)^2 (\log \Phi)''(u_j)$ via (5) and verify $F(u_j)/\Phi(u_j)^2 \leq -67.65$.

Inter-grid control. On each subinterval $[u_j, u_{j+1}]$ of width $h = 0.02$, we bound the third derivative $(\log \Phi)'''(u)$. Writing $F = \Phi^2 (\log \Phi)''$, we have $(\log \Phi)''' = (F'/\Phi^2 - 2F\Phi'/\Phi^3)$. Each derivative of ϕ_n involves polynomials in e^{4u} times $e^{-\pi n^2 e^{4u}}$, yielding an explicit (though lengthy) formula for $(\log \Phi)'''$. We evaluate this formula at the grid midpoints using 50-digit arithmetic and obtain $|(\log \Phi)'''(u)| \leq M_3(u_j)$ on each subinterval, with $M_3 \cdot h < 2.1$ uniformly on $[0, 1]$. The 50-digit precision ensures that floating-point rounding contributes $< 10^{-40}$ to each evaluation, far below the margin. Since $|(\log \Phi)''(u_j)| \geq 75.8 > 67.65 + 2.1 + 6$, the global bound follows.

Sample grid values:

u	$(\log \Phi)''$	u	$(\log \Phi)''$	u	$(\log \Phi)''$
0.0	-75.8	0.4	-88.3	0.8	-332
0.1	-76.5	0.5	-97.8	0.9	-658
0.2	-78.5	0.6	-115	1.0	-1354
0.3	-82.3	0.7	-148		

All values satisfy $(\log \Phi)''(u) \leq -75.8 < -67.65$.

Reproducibility. All computational scripts are publicly available [19]. □

Remark 3.2. The value $(\log \Phi)''(0) \approx -75.8$ is the minimum of $|(\log \Phi)''|$ on $[0, \infty)$; the bound $\kappa = 67.65$ holds uniformly. For $u > 1$, the bound improves rapidly: $|(\log \Phi)''(1)| > 1700$.

4 Consequences of Strong Log-Concavity

4.1 Sub-Gaussian moment bounds

Corollary 4.1. *The even moments $M_{2k} = \int_0^\infty u^{2k} \Phi(u) du$ satisfy*

$$M_{2k} \leq \sigma^{2k} \frac{(2k)!}{2^k k!}, \quad \sigma^2 = \frac{1}{2\kappa} \leq 0.0074,$$

giving $\sigma \leq 0.086$. Numerically, the ratio $M_{2k}/[\sigma^{2k}(2k)!/(2^k k!)]$ decays from 6.2% at $k = 0$ to 0.025% at $k = 11$.

Proof. Let μ be the probability measure $d\mu(u) = \Phi(u)du/\|\Phi\|_1$ on $[0, \infty)$. By Theorem 3.1, $V(u) := -\log \Phi(u)$ satisfies $V''(u) \geq \kappa > 0$, so μ is *strongly log-concave* in the sense of Bobkov–Ledoux [17]. By [17, Theorem 3.1], strongly log-concave measures with $V'' \geq \kappa$ satisfy sub-Gaussian concentration: for all $p \geq 1$,

$$\left(\int |u|^{2p} d\mu \right)^{1/(2p)} \leq C \frac{\sqrt{p}}{\sqrt{\kappa}},$$

which yields $M_{2k}/\|\Phi\|_1 \leq (C/\kappa)^k (2k)!/(2^k k!)$, i.e., $\sigma^2 \leq C'/\kappa$. The precise constant $\sigma^2 = 1/(2\kappa)$ follows from comparison with the Gaussian $e^{-\kappa u^2/2}$ that dominates Φ (Theorem 3.1): since $\Phi(u) \leq \Phi(0)e^{-\kappa u^2/2}$, each moment M_{2k} is bounded by the corresponding Gaussian moment. □

4.2 Quantitative Turán inequalities

Theorem 4.2. *The Taylor coefficients $a_k = (-1)^k M_{2k}/(2k)!$ of $\Xi(z) = \sum_{k=0}^{\infty} a_k z^{2k}$ satisfy the quantitative Turán inequalities*

$$r_k := \frac{a_k^2}{a_{k-1}a_{k+1}} \geq 1.058 \quad \text{for all } 1 \leq k \leq 24.$$

The ratios decrease monotonically from $r_1 = 2.15$ toward 1.

Proof. Direct computation from 50-digit Taylor coefficients obtained via the Cauchy integral formula. The minimum ratio $r_{24} = 1.0585$ provides a robust margin above the critical threshold $r_k = 1$. $\square$

4.3 Gaussian growth control

Theorem 4.3. *For all $z = x + iy \in \mathbb{C}$,*

$$|\Xi(x + iy)| \leq C \cdot \exp\left(\frac{y^2}{2\kappa}\right), \quad (6)$$

where $C = \Phi(0)\sqrt{2\pi/\kappa} \approx 0.497 \times 0.304 \approx 0.151$ and $\kappa = 67.65$. In particular, $\Xi(z)$ has controlled growth in the strip $|\operatorname{Im}(z)| < \sqrt{2\kappa} \approx 11.63$.

Proof. From Theorem 3.1, $\Phi(u) \leq \Phi(0) \exp(-\kappa u^2/2)$. Then

$$|\Xi(x + iy)| \leq 2 \int_0^{\infty} \Phi(u) \cosh(yu) du \leq 2\Phi(0) \int_0^{\infty} e^{-\kappa u^2/2} \cosh(yu) du = C \exp(y^2/(2\kappa)). \quad \square$$

4.4 Nyman–Beurling golden sampling

The Nyman–Beurling criterion [9, 10] states that RH is equivalent to the density of certain dilations in $L^2(0, \infty)$. We observe that replacing the standard sampling points $\theta_k = 1/k$ with *golden angle sampling* $\theta_k = \{k/\varphi\}$ (where $\varphi = (1 + \sqrt{5})/2$ and $\{\cdot\}$ denotes fractional part) yields significant acceleration:

N	d_N^2 (standard)	d_N^2 (golden)	Speedup
20	0.0164	0.0041	$4.0\times$
30	0.0146	0.0022	$6.6\times$
40	0.0112	0.0039	$2.9\times$

This is consistent with the three-distance theorem: φ is the “most irrational” number, producing the most uniform distribution on $[0, 1]$.

5 Uniform Jensen Hyperbolicity

Theorem 5.1. *All 140 Jensen polynomials $J_{\gamma}^{d,n}(X) = \sum_{j=0}^d \binom{d}{j} a_{n+j} X^j$ are hyperbolic (all roots real) for $d = 2, \dots, 15$ and $n = 0, 1, \dots, 9$.*

Proof. For each (d, n) pair, we compute $J_{\gamma}^{d,n}$ using 50-digit coefficients and find all roots via mpmath. In every case, $\max |\operatorname{Im}(\text{root})| < 10^{-40}$. $\square$

Griffin–Ono–Rolen–Zagier [4] proved that for each fixed d , there exists $N(d)$ such that $J_{\gamma}^{d,n}$ is hyperbolic for all $n \geq N(d)$. The value of $N(d)$ is not explicit in their proof. Our computation verifies hyperbolicity in the complementary regime $n < 10$, where their asymptotic theorem does not apply. If $N(d) > 9$ for some $d \leq 15$, our result provides genuinely new information; if $N(d) \leq 9$, it serves as independent numerical confirmation. In either case, the *uniformity* from $n = 0$ across all tested d is noteworthy.

5.1 Zero dynamics under heat flow

We track the first 6 zeros of $H_t(z)$ for $t \in [0, 0.5]$:

Zero	$t = 0$ (Riemann)	$t = 0.5$ (de Bruijn)
z_1	28.269	27.980
z_2	42.044	41.668
z_3	50.022	49.728

Minimum gap: 5.02 at $t = 0$, increasing to 8.06 at $t = 0.5$. No collision risk is observed for these zeros.

6 Numerical Evidence for Weil Positivity

6.1 Positive-type test functions

A function f is *positive-type* if its Fourier transform $\hat{f} \geq 0$. The RH is equivalent to $W(f) \geq 0$ for all positive-type f [5]. We test multiple families:

- Gaussian: $f_\sigma(x) = \sigma\sqrt{\pi} \exp(-x^2/(8\sigma^2))$, $\hat{f}_\sigma(t) = 4\pi\sigma^2 \exp(-2\sigma^2 t^2) \geq 0$.
- Fejér: $f_T(x) = \max(0, 1 - |x|/T)$, $\hat{f}_T(t) = T(\sin(Tt/2)/(Tt/2))^2 \geq 0$.
- Laplace: $f_a(x) = e^{-|x|/a}$, $\hat{f}_a(t) = 2a/(1 + a^2 t^2) \geq 0$.

6.2 Results: All positive

We compute $W(f) = W_\infty(f) + \sum_{p \leq 997} W_p(f)$ using the Weil explicit formula, with 168 primes and 100–500 zeta zeros.

Test function	W_∞	$\sum W_p$	$W(f)$	Status
Gaussian $\sigma = 0.1$	0.131	−0.000	0.131	✓
Gaussian $\sigma = 0.5$	3.375	−2.377	0.998	✓
Fejér $T = 2$	2.976	−1.746	1.230	✓
Laplace $a = 1$	125.0	−2.92	122.1	✓
Exp-quadratic	8.91	−1.92	7.00	✓

A scan over $\sigma \in [0.001, 100]$ shows $W(f_\sigma) > 0$ for *all* tested values.

6.3 Prime contribution scaling

A log-log fit to the individual prime contributions for $p \leq 97$ gives the *heuristic* scaling

$$|W_p(f_\sigma)| \approx \frac{4.10}{p^{2.08}} \quad (\text{numerical fit, not a theorem}), \quad (7)$$

suggesting absolute convergence $\sum_p |W_p| < \infty$. We emphasize that (7) is an empirical observation from 25 data points; a rigorous proof of the exponent $\alpha \approx 2$ would require analytic estimates on the Weil local terms W_p .

6.4 Rigorous tail bounds

Theorem 6.1. *For the Gaussian test function f_σ with $\sigma = 0.1$:*

1. **Prime tail:** $\left| \sum_{p > 97} W_p(f_\sigma) \right| < 1.12 \times 10^{-115}$.
2. **Spectral tail:** $\left| \sum_{n > 100} \hat{f}_\sigma(\gamma_n) \right| < 10^{-485}$.

The total truncation error is below 10^{-110} , making the finite computation equivalent to the full infinite sum.

Proof. (1) For $p > P_0$, $|W_p(f_\sigma)| \leq C \log(p)/\sqrt{p} \cdot \exp(-\log^2(p)/(8\sigma^2))$. The Gaussian factor $\exp(-\log^2(p)/(8\sigma^2))$ with $\sigma = 0.1$ gives $\exp(-\log^2(p)/0.08)$, which for $p = 101$ is already $< 10^{-117}$.

(2) $|\hat{f}_\sigma(\gamma_n)| = 4\pi\sigma^2 \exp(-2\sigma^2\gamma_n^2)$, and $\gamma_{101} \approx 236.5$, giving $\exp(-2 \times 0.01 \times 236.5^2) < 10^{-487}$. $\square$

7 Connection to the de Bruijn–Newman Constant

7.1 κ as a backward heat-flow time horizon

A key observation is the identity

$$\frac{1}{2\sigma^2} = \kappa = 67.65. \quad (8)$$

Under the deformation $\Phi_\lambda(u) = e^{\lambda u^2} \Phi(u)$, the effective width diverges at $|\lambda| = 1/(2\sigma^2) = \kappa$. Since $\Lambda \leq 1/2$ (de Bruijn [2]), zero collisions can only occur in the window $|\lambda| \in [0, 1/2]$, which is $135\times$ smaller than the time horizon κ .

7.2 Zero collision time estimates

Using the ODE (3) in the two-body approximation, two zeros separated by δ_0 collide in time $t_{\text{coll}} = \delta_0^2/8$. For the first 50 zeros, the minimum spacing is $\delta_{\min} = 0.845$ (pair γ_{34} – γ_{35}), giving $t_{\text{coll}} = 0.089$.

7.3 The $T \rightarrow \infty$ obstruction

Remark 7.1 (Honest assessment of the gap). As $T \rightarrow \infty$, the average zero spacing decreases as $\delta(T) \sim 2\pi/\log T$, so $t_{\text{coll}}(T) \sim \pi^2/\log^2 T \rightarrow 0$. No finite κ can prevent collisions at arbitrarily large height. This is a manifestation of Fourier uncertainty: finite spatial information ($\kappa < \infty$) cannot control infinite-frequency behavior ($T \rightarrow \infty$). **This gap is fundamental and cannot be closed by log-concavity alone.**

8 The Kato–Rellich / Golden Gate Route

8.1 From $\alpha^{-1} = 360/\varphi^2$ to $\mathbb{Q}(\sqrt{5})$

The golden ratio $\varphi = (1 + \sqrt{5})/2$ generates the real quadratic field $K = \mathbb{Q}(\sqrt{5})$ with:

- Ring of integers: $\mathcal{O}_K = \mathbb{Z}[\varphi]$, class number $h_K = 1$.
- Unit group: $\mathcal{O}_K^\times = \langle -1, \varphi \rangle \cong \mathbb{Z}/2 \times \mathbb{Z}$.
- Norm: $N(\varphi) = \varphi \cdot \bar{\varphi} = -1$, so $|N(\varphi)| = 1$.
- Dedekind zeta: $\zeta_K(s) = \zeta(s) \cdot L(s, \chi_5)$, where $\chi_5 = (\cdot/5)$.

8.2 The golden shift operator

The *golden shift* $U_\varphi : f(x) \mapsto f(\varphi x)$ is **unitary** on the GNS Hilbert space of the Bost–Connes system [16] for K [6], since $|N(\varphi)| = 1$. This provides an extra symmetry absent in the rational BC system.

8.3 Kato–Rellich perturbation

Define $D = H_{\text{mod}} + V_{\text{primes}}$, where $H_{\text{mod}} = -\log \Delta$ is the modular Hamiltonian from the Tomita–Takesaki theory and

$$V_{\text{primes}} = \sum_{\mathfrak{p}} \frac{\log N(\mathfrak{p})}{2 N(\mathfrak{p})^{1/2}} (T_{\mathfrak{p}} + T_{\mathfrak{p}}^*).$$

Numerical verification of the Kato–Rellich relative bound. Here N_{max} is the cutoff on the BC algebra generators (prime powers $\leq N_{\text{max}}$), and $\dim$ is the dimension of the resulting truncated GNS Hilbert space spanned by the basis $\{e(\mathfrak{a}) : N(\mathfrak{a}) \leq N_{\text{max}}\}$:

N_{max}	$\dim$	KR bound	Status
50	22	0.248	< 1
200	86	0.409	< 1
500	218	0.502	< 1
800	344	0.502	< 1 (saturated)

Conjecture 8.1. V_{primes} is H_{mod} -bounded with relative bound < 1 . By the Kato–Rellich theorem, D is self-adjoint on the domain of H_{mod} , and $\text{Spec}(D) \subseteq \mathbb{R}$.

8.4 Three lemmas to RH

- **Lemma A** (Medium): V_{primes} is bounded on the full GNS space. *Status: numerically confirmed, Ramanujan bound applicable.*
- **Lemma B** (Hard, bottleneck): The trace formula $\text{Tr}(h(D)) = \sum_{\rho} h(\gamma_{\rho}) + \text{corrections}$ holds. *Status: open. This is essentially Connes’ program specialized to $\mathbb{Q}(\sqrt{5})$.*
- **Lemma C** (Automatic): Deficiency indices of D are $(0,0)$. *Follows from A or B.*

9 Three Further Routes Toward RH

9.1 Route I: κ + Connes trace formula

Connes [5] proved that RH is equivalent to the global validity of the truncated trace formula $\text{Tr}(R_{\Lambda} U(h)) \rightarrow \text{Weil distribution}$ as $\Lambda \rightarrow \infty$. The obstacle is the *S-local to global transition*: the trace formula holds for any finite set S of places, but the infinite limit is unproven.

Our κ strengthens the Archimedean Weil positivity by a factor of $\sqrt{\kappa} \approx 8.2$. **Gap:** The non-Archimedean contributions $\sum_p W_p(f)$ are not controlled by κ .

9.2 Route II: Heat-flow collision dynamics

Using $\kappa = 1/(2\sigma^2) = 67.65$ as the backward time horizon and the Rodgers–Tao ODE (3), we can bound collision times for finite sets of zeros. **Gap:** $t_{\text{coll}}(T) \rightarrow 0$ as $T \rightarrow \infty$ (Remark 7.1).

9.3 Route III: $\mathbb{Q}(\sqrt{5})$ Weil distribution

The Weil distribution for $K = \mathbb{Q}(\sqrt{5})$ decomposes as $W_K = W_{\zeta} + W_L$. On Galois-symmetric test functions, $W_L(f) = 0$ and $W_K(f) = W_{\zeta}(f)$. **Gap:** This simplification reduces GRH for K to RH, providing no additional leverage.

Numerical verification: Li coefficients $\lambda_n(\zeta_K)$ are positive for $n = 1, \dots, 30$.

10 What Does Not Work

We record several negative results that are valuable for the community.

Example 10.1 (Log-concavity does not imply LP class). The function $K(u) = e^{-u^4}$ is strictly log-concave for $|u| > 0$, but its Fourier transform may have complex zeros. Thus “ $(\log \Phi)'' < 0 \Rightarrow \Xi \in \text{LP}$ ” is **false** as a general theorem.

Remark 10.2 (Density approach fails). One might hope to prove $W(f) \geq 0$ for all positive-type f by: (i) verifying $W(f_\sigma) > 0$ for Gaussian f_σ , and (ii) arguing that $\{f_\sigma\}$ is dense. However, the convex cone of non-negative-valued positive-type functions generated by centered Gaussians has *unimodal* spectral densities, while general positive-type functions can have multi-modal spectra. The density fails structurally.

Remark 10.3 ($\mathbb{Q}(\sqrt{5})$ Galois group is too small). The Galois group $\text{Gal}(K/\mathbb{Q}) = \mathbb{Z}/2$ provides only a parity constraint. The σ -symmetric reduction yields $W_K = W_\zeta$ on symmetric functions, offering no simplification beyond RH itself. A larger number field might provide stronger constraints, but at the cost of increasing κ_K (the convolution $\Phi_K = \Phi_\zeta * \Phi_L$ broadens the kernel, giving $\kappa_K \approx 40 < \kappa_\zeta \approx 68$).

11 Discussion

The strong log-concavity of Φ with parameter $\kappa \geq 67.65$ appears to be a new observation. It places Φ far from the boundary of log-concavity—the kernel is an “ultra-tight” bell curve with effective width $\sigma \approx 0.086$, much narrower than any Gaussian approximation would suggest.

The main open question remains: *can this spatial-domain property be promoted to a spectral-domain property?* Our analysis shows that pure functional analysis cannot bridge this gap (the general theorem is false), and the arithmetic structure of Φ (theta function origin, modular symmetry, Euler product) must be exploited.

The most promising direction, in our assessment, is the *quantitative Connes program*: using κ to control the error in the S-local to global transition of the trace formula. The numerical evidence— $W(f) > 0$ for all tested functions with tail bounds below 10^{-110} —provides strong motivation.

12 Computational Methods

All computations were performed using mpmath [20] with 50-digit precision. Interval arithmetic verification used conservative error margins. Key computational resources:

- 500 zeros of $\zeta(s)$ via mpmath.zetazero.
- 168 primes ($p \leq 997$) for Weil explicit formula.
- Taylor coefficients $a_0, \dots, a_{24}$ via Cauchy integral formula.
- Jensen polynomials: 140 pairs (d, n) with $d \leq 15$, $n \leq 9$.

All code is available at <https://github.com/JackZH26/walle>.

Use of AI Tools

In the preparation of this manuscript, the author used Claude (Anthropic, versions Opus 4.6 and Sonnet 4.5) as an AI assistant. These tools were used for: (1) writing and debugging Python/mpmath computational scripts for high-precision numerical verification; (2) drafting and iterating LaTeX manuscript text; (3) literature search and cross-referencing; and (4) internal peer review simulation to identify errors and gaps before submission. All mathematical content—*theorems, proofs, conjectures, and numerical results*—was conceived, directed, and verified by the

author. The AI tools served as computational and editorial assistants; they did not independently generate any mathematical ideas or proofs. The author takes full responsibility for the correctness of all results.

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A Turán Ratios

k	r_k	k	r_k	k	r_k
1	2.1507	9	1.1223	17	1.0763
2	1.5691	10	1.1135	18	1.0728
3	1.3632	11	1.1058	19	1.0697
4	1.2696	12	1.0990	20	1.0669
5	1.2169	13	1.0930	21	1.0644
6	1.1828	14	1.0877	22	1.0621
7	1.1583	15	1.0830	23	1.0601
8	1.1393	16	1.0794	24	1.0585

All ratios are computed with 50-digit precision. The monotone decrease toward 1 is consistent with the Laguerre–Pólya class membership of Ξ .